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DENSITY-ADAPTIVE GRAPH LAPLACIAN CONSTRUCTION FOR ROBUST SPECTRAL CLUSTERING

Deep Feature Based Land-Use Image Clustering using ResNet Embeddings

Group -C11

TEAM MATES :

- | | |
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INTRODUCTION :

Many satellite images are collected every day, and organizing them manually is difficult. Clustering helps automatically group similar images. However, simple clustering methods do not work well for complex image data. In this project, we use spectral clustering and improve it using a density-adaptive graph method to achieve better clustering performance.



LITERATURE REVIEW :

Authors & Year	Paper Title	Core Idea	Methodology	Limitations Identified	Relevance to Our Project
Ng, Jordan, Weiss (2001)	On Spectral Clustering: Analysis and an Algorithm	Formalized spectral clustering using graph Laplacians and eigenvectors	Construct affinity matrix using global Gaussian kernel, normalize Laplacian, cluster eigenvectors using K-means	Uses a single global similarity scale (σ), sensitive to parameter choice, assumes uniform data density	Provides the theoretical foundation for spectral clustering and motivates why improving graph construction is necessary
Ulrike von Luxburg (2007)	A Tutorial on Spectral Clustering	Theoretical and intuitive understanding of spectral clustering	• Formulated clustering as a graph partitioning problem • Defined unnormalized and normalized graph Laplacians • Analyzed spectral properties and eigenvector behavior • Derived spectral clustering via graph cuts, random walks, and perturbation theory	• Highlights strong dependence on similarity graph construction • Does not propose adaptive similarity scaling • Notes lack of theory on optimal graph design	Provides the theoretical foundation for graph Laplacians and motivates operator-level analysis of similarity graphs
Zelnik-Manor & Perona (2004)	Self-Tuning Spectral Clustering	Sensitivity of spectral clustering to global scale parameter σ and inability to handle multi-scale data	• Introduced local (density-adaptive) scaling • Removed dependence on a single global σ • Improved clustering in multi-scale and cluttered data	• Defined local scale σ_i using k-th nearest neighbor distance • Constructed adaptive affinity: $A_{ij} = \exp(-\ x_i - x_j\ ^2 / (\sigma_i \sigma_j))$ • Built normalized Laplacian and analyzed eigenvectors	• Focuses mainly on clustering quality • Limited operator-level spectral analysis • Does not deeply study eigenvalue stability

PROBLEM STATEMENT

- Large numbers of satellite images are generated every day.
- Manually organizing and labeling these images is difficult and time-consuming.
- Traditional clustering methods like K-Means often fail to handle complex image data.
- Spectral clustering works better, but its performance depends on how the similarity graph is constructed.
- Using a single global similarity parameter may not work well when data has different density regions.

“Therefore, there is a need for a density-adaptive graph construction method to improve the robustness and accuracy of spectral clustering.”

PROJECT VISION AND MISSION :

PROJECT VISION :

- To explore how graph construction influences the spectral properties of data beyond standard clustering pipelines.
- To demonstrate that incorporating local density information leads to more robust and stable spectral representations.
- To bridge theoretical concepts of graph Laplacians with practical applications in real-world image datasets.
- To move beyond syllabus-level implementation by focusing on operator-level design and analysis.

PROJECT MISSION :

- To use deep learning features to represent satellite images.
- To apply spectral clustering for grouping similar images.
- To improve clustering performance using a density-adaptive graph method.
- To evaluate the effectiveness of the clustering results using standard metrics.

DATASET DESCRIPTION:

Dataset Name :

The dataset used in this project is the EuroSAT Dataset.

EuroSAT is a satellite image dataset for land-use and land-cover classification derived from Sentinel-2 satellite imagery.

The EuroSAT dataset contains satellite images of different land-use categories.

Original dataset characteristics:

Total images: 27,000

Number of classes: 10

Image size: 64×64 pixels

Image type: RGB satellite images

Each image represents a specific land-use class.

We selected five classes from the dataset:

- Highway
- Residential
- Forest
- River
- SeaLake

These classes were selected because they represent distinct land-use patterns and provide a good clustering challenge.

DATASET DESCRIPTION:

Dataset Size Used :

To maintain a balanced dataset, we selected:

400 images per class

Therefore,

Total Images= $5 \times 400 = 2000$

2000 satellite images

5 land-use classes

balanced distribution

Balanced datasets help prevent bias in clustering algorithms.

Image Properties :

Each image has the following characteristics:

Resolution: 64×64 pixels

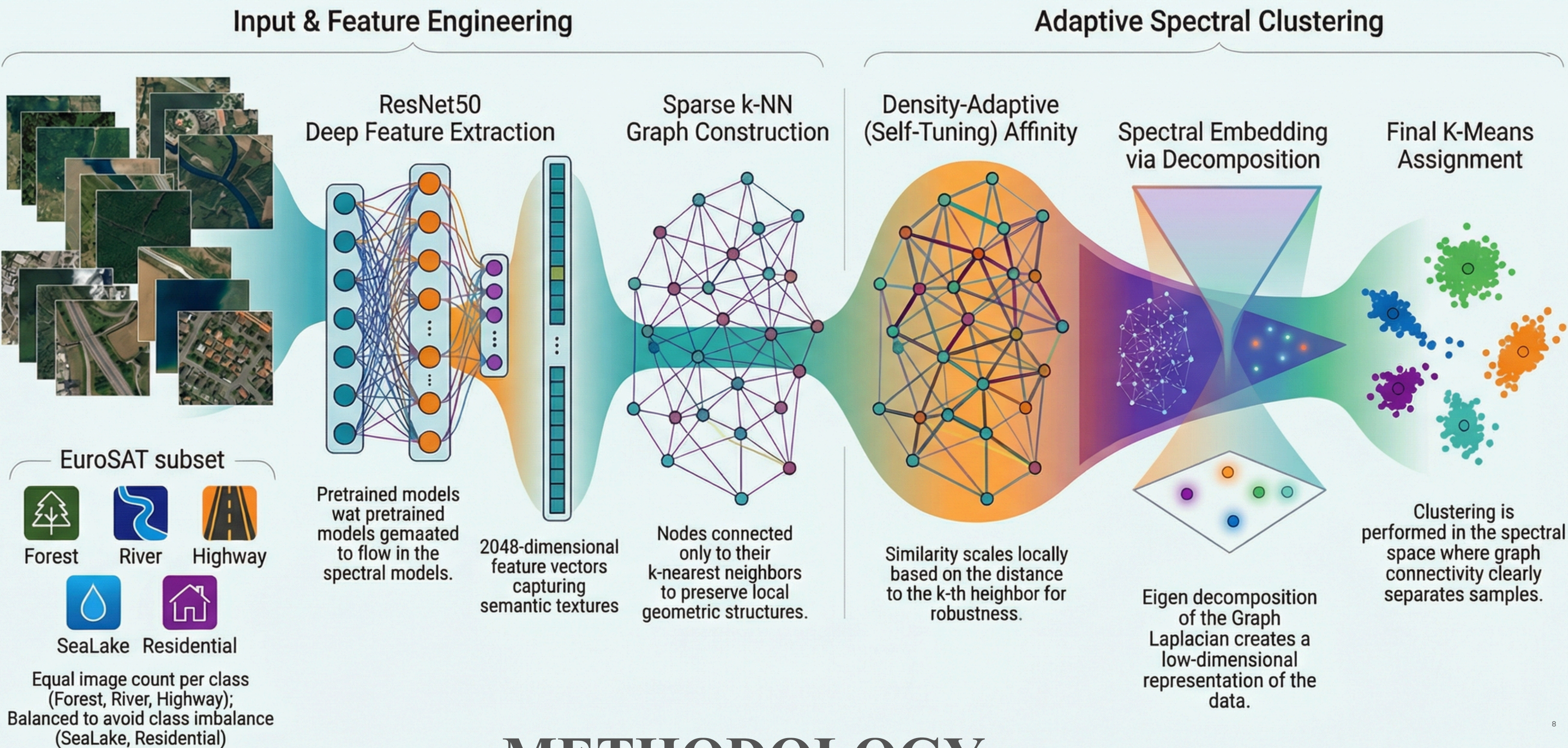
Color channels: RGB (3 channels)

Image format: JPEG

Data type: satellite imagery

These images represent top-down views of Earth's surface captured by satellites.

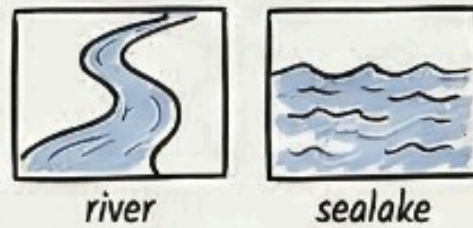
Mapping with Precision: The Density-Adaptive Spectral Clustering Workflow



Mapping the Earth: A Deep Learning Approach to Satellite Image Clustering

1. Data Preparation & Preprocessing

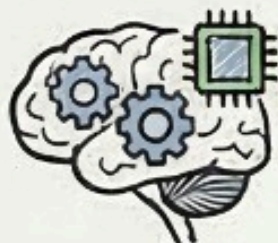
EuroSAT Dataset (2,000 images, 5 classes, 400 each):



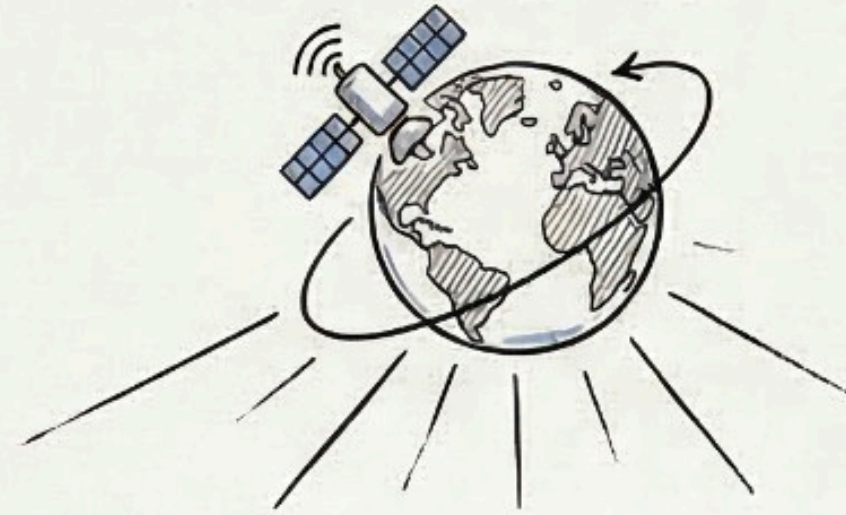
EuroSAT Dataset
(2,000 images, 5 classes, 400 each):



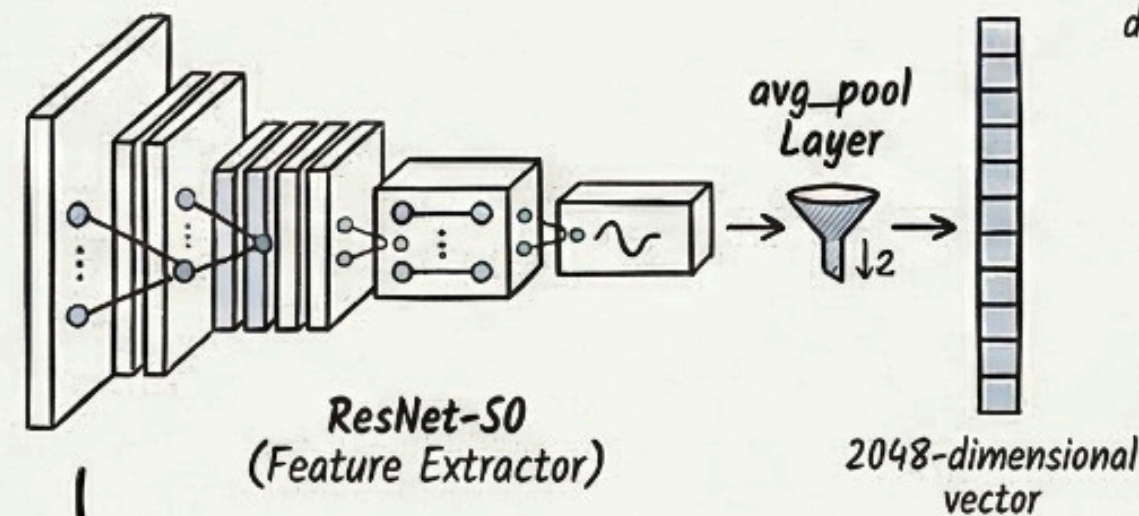
224 × 224 × 3 pixels
Resizing for ResNet



Memory Efficiency:
Batch processing via datastore
reduces memory usage.



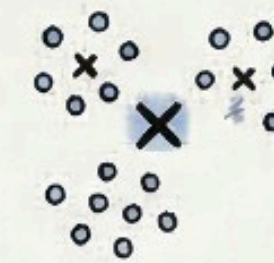
2. Deep Feature Extraction



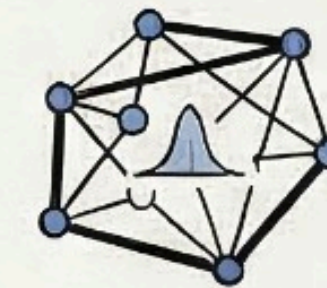
	1	2	3	4	...	...
1						
2						
3						
4						
5						
6						

The Feature Matrix (X):
2000 images × 2048 features
(mathematical fingerprint)

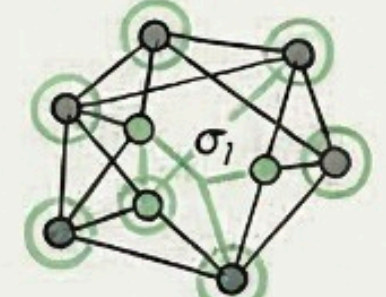
3. The Clustering Algorithms



K-Means
(The Baseline)
Groups by Euclidean
distance to centroids

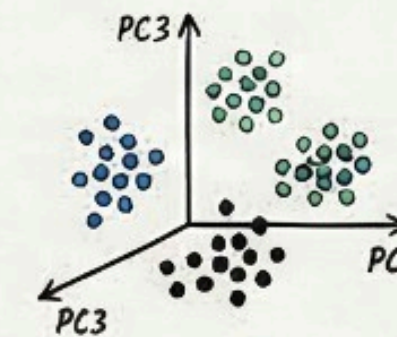


Spectral Clustering
(Global Sigma)
Treats dataset as a
graph, uses Gaussian
kernel for similarity



Self-Tuning
Spectral Clustering
Calculates local scaling (σ_i)
based on 25 nearest
neighbors, adapts to density

4. Evaluation & Visualization



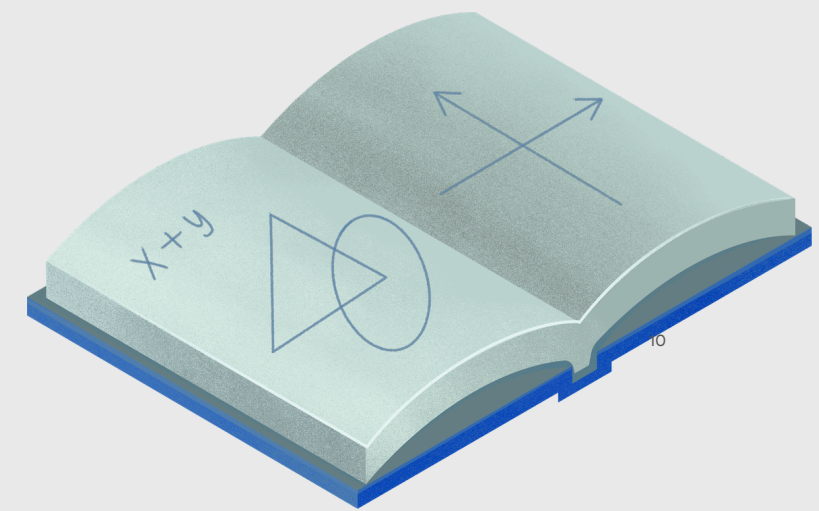
Dimensionality Reduction (PCA)
Compresses 2048 features into
2D/3D "maps" for visualization

Performance Metrics (Accuracy - ACC)	
K-Means:	0.8570
Spectral (Global Sigma):	0.8195
Self-Tuning Spectral:	0.9035

Success measured via Accuracy (ACC), NMI, and ARI.

-METHODOLOGY-

MATHEMATICAL FRAMEWORK



1. Deep Feature Representation

Each satellite image is transformed into a numerical feature vector using pretrained convolutional neural network (CNN) models such as ResNet18, ResNet50, and ResNet101.

Let the dataset contain n satellite images:

Dataset Representation : $I = I_1, I_2, I_3, \dots, I_n$

where

- I_i represents the i -th satellite image
- n represents the total number of images

Feature Extraction using Multiple CNN Models :

Each CNN model extracts deep features from the images.

Feature Extraction Function : $x_i(m) = f_m(I_i)$

where :

$f_m(\cdot)$ represents the feature extraction function of the m -th CNN model .

$x_i^{(m)}$ represents the feature vector of the i -th image extracted using model m

2. Feature Matrix Representation

After extracting deep features from all satellite images using the CNN models, each image is represented by a feature vector. These feature vectors capture important patterns in the images such as texture, shape, spatial structure, and land-use characteristics.

Suppose the dataset contains n images and a CNN model extracts d -dimensional features for each image.

For the m -th CNN model, the extracted features for all images can be organized into a feature matrix.

The feature matrix is written as:

$$\mathbf{X}^m = [\mathbf{x}_1^m, \mathbf{x}_2^m, \mathbf{x}_3^m, \dots, \mathbf{x}_n^m]^T$$

Where:

- $\mathbf{x}_i^m$ represents the feature vector extracted from the i -th image using the m -th CNN model
- m represents the index of the CNN model used for feature extraction (for example ResNet18, ResNet50, ResNet101, etc.)
- n represents the total number of images in the dataset

Each feature vector has d elements, therefore:

$$\mathbf{x}_i^m = [f_1, f_2, f_3, \dots, f_d]$$

Where:

- $f_1, f_2, \dots, f_d$ represent the extracted deep features describing the image.

3 .Similarity Matrix Construction

After extracting deep features from the satellite images, the dataset is represented as a feature matrix. However, spectral clustering does not directly cluster the feature vectors. Instead, it constructs a similarity graph that represents relationships between images.

To build this graph, we compute a similarity matrix, which measures how similar every pair of images is.

Definition of Similarity Matrix

The similarity matrix is denoted as W .

Each element of the matrix represents the similarity between two images.

$$W = [W_{ij}]$$

Where:

- W_{ij} represents the similarity between image i and image j

If two images are very similar:

$$W_{ij} \rightarrow \text{large value}$$

If two images are very different:

$$W_{ij} \rightarrow \text{small value}$$

The similarity matrix has size:

$$W \in \mathbb{R}^{n \times n}$$

Where n is the number of images.

Thus the matrix compares every image with every other image.

Gaussian Similarity Function :

To measure similarity between feature vectors, the Gaussian kernel (Radial Basis Function) is used.

The similarity between two feature vectors $\mathbf{x}_i$ and $\mathbf{x}_j$ is defined as:

$$\mathbf{W}_{ij} = \exp(- \|\mathbf{x}_i - \mathbf{x}_j\|^2 / (2\sigma^2))$$

Where:

- $\|\mathbf{x}_i - \mathbf{x}_j\|^2$ represents the squared Euclidean distance between feature vectors
- σ represents the scaling parameter controlling similarity

Interpretation of Gaussian Similarity :

The Gaussian kernel converts distance into similarity.

Case 1 — Images are very similar

distance $\rightarrow$ small

similarity $\rightarrow$ close to 1

Case 2 — Images are different

distance $\rightarrow$ large

similarity $\rightarrow$ close to 0

This means that images with similar visual features will have strong connections in the graph.

4. K-Means Clustering

Objective of K-Means :

K-Means partitions the dataset into K clusters by minimizing the distance between data points and cluster centers.

Objective Function :

$$J = \sum_{i=1}^n \sum_{j=1}^K \|x_i - \mu_j\|^2$$

Where:

- x_i = feature vector of data point i
- μ_j = centroid of cluster j

Algorithm Steps :

- Initialize K cluster centroids randomly
- Assign each data point to the nearest centroid

Update cluster centroids using:

$$\mu_j = \frac{1}{|C_j|} \sum_{x_i \in C_j} x_i$$

Repeat until cluster assignments stabilize

THE CORE IDEA :

Density-Adaptive Similarity

To overcome this limitation, a density-adaptive similarity measure is used.
Instead of using a single global σ , each data point has its own local scale parameter.

The similarity becomes:

$$\mathbf{W}_{ij} = \exp(- \| \mathbf{x}_i - \mathbf{x}_j \|^2 / (\sigma_i \sigma_j))$$

Where:

σ_i = distance between point $\mathbf{x}_i$ and its k -th nearest neighbor

This means the similarity adapts based on local data density.

Spectral Clustering Foundation :

1. Degree Matrix

The degree matrix represents the total similarity connected to each node.

Equation

$$D_{ii} = \sum W_{ij}$$

Where :

- W_{ij} = similarity between nodes i and j

Properties :

- D is a diagonal matrix
- Each value represents the total similarity weight of a node

2. Graph Laplacian

The normalized graph Laplacian captures the structure of the similarity graph.

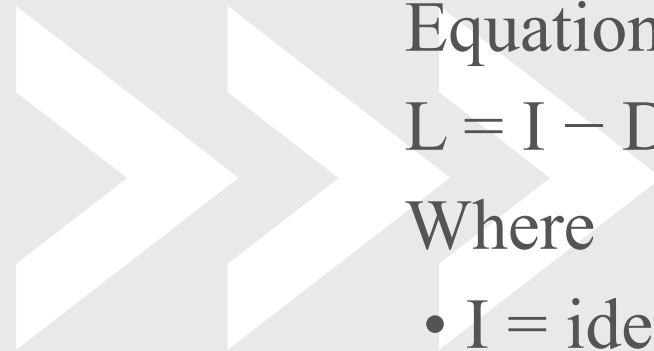
Equation :

$$L = I - D^{-1/2} W D^{-1/2}$$

Where

- I = identity matrix
- W = similarity matrix
- D = degree matrix

The Laplacian helps reveal cluster structure in the graph.



Spectral Clustering Foundation :

3. Spectral Decomposition :

Eigenvectors of the Laplacian are computed.

Equation :

$$L u = \lambda u$$

The first k eigenvectors are selected to form the spectral embedding.

Spectral Representation :

$$U = [u_1, u_2, \dots, u_k]$$

Each row of U represents a data point in spectral space.

4. Clustering :

Finally, K-Means clustering is applied to the spectral embeddings.

Objective Function :

$$\sum \|U_i - C_j\|^2$$

Where :

- U_i = spectral representation of data point i
- C_j = cluster centroid

Each data point is assigned to the nearest cluster center.

Results & Explanation

1) KMeans Cluster assignment matrix

True Class \ Predicted Class	1	2	3	4	5
1		394		5	1
2		1	299	96	4
3		6	16	2	376
4		2	112	285	1
5	360	6	1	33	

2) Spectral (Global Sigma) Cluster assignment matrix

True Class \ Predicted Class	1	2	3	4	5
1	44			356	
2	80		305	5	10
3	7		18	4	371
4	252	1	143	3	1
5	35	355	4	6	

True Class \ Predicted Class	1	2	3	4	5
1	395		1	4	
2	2		6	392	
3	1		399		
4	2		1	396	1
5	6	225		20	149

3) Spectral (Self-Tuning) Cluster assignment matrix

1) KMeans Cluster assignment matrix

- Class 1 mostly correct (394); Class 3 $\rightarrow$ Cluster 4 (376); Class 5 $\rightarrow$ Cluster 1 (360).
- Minor confusion between River–SeaLake and Forest–Residential due to similar textures in deep features; overall clustering still strong.

2) Spectral (Global Sigma) Cluster assignment matrix

Many samples are misclustered (e.g., Class 1 $\rightarrow$ C4, Class 4 $\rightarrow$ C1, Class 5 $\rightarrow$ C2) because global σ over-smooths the graph, creating excessive long-range connections and reducing cluster separability.

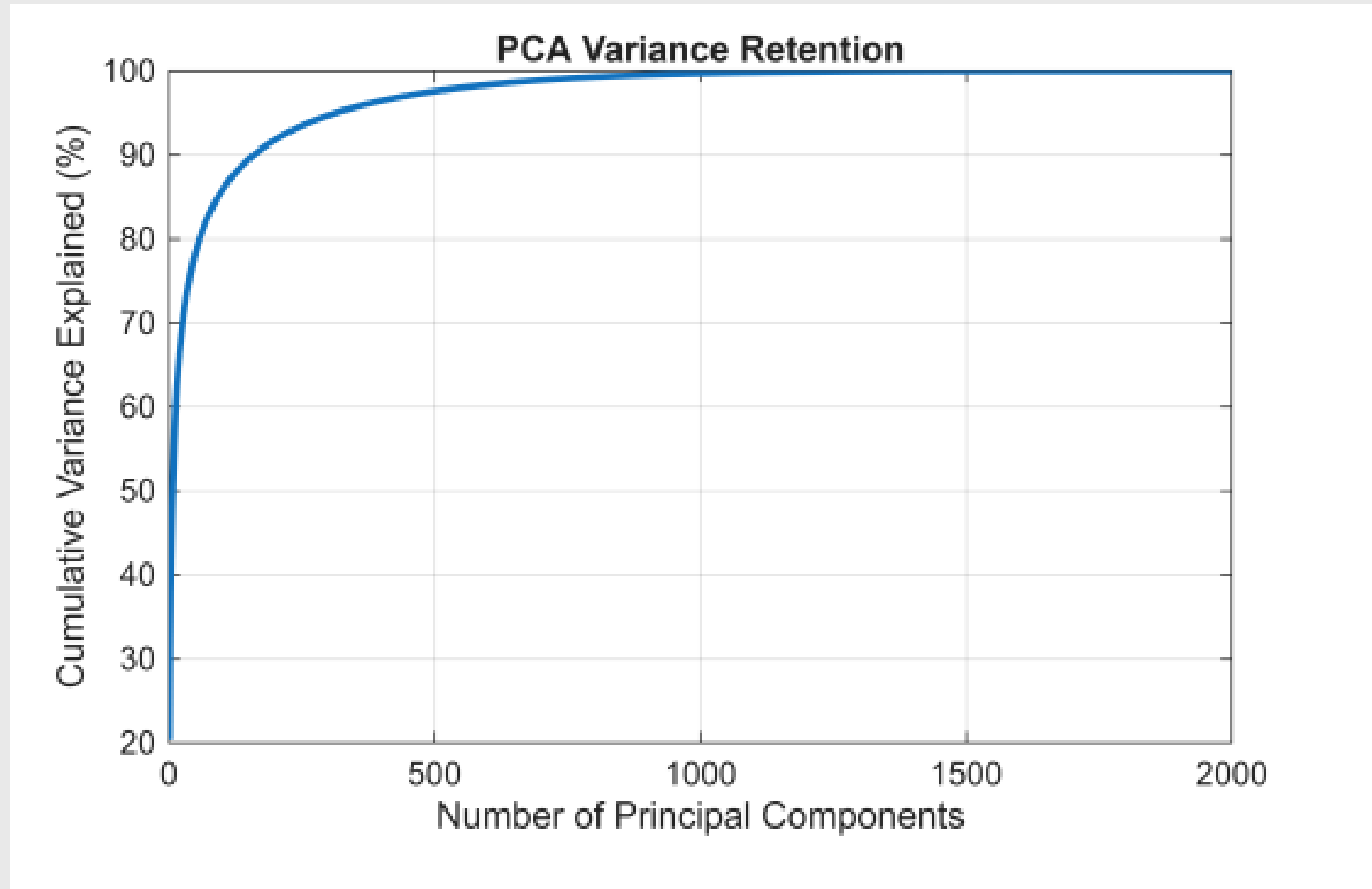
3) Spectral (Self-Tuning) Cluster assignment matrix

- Classes 1, 3, and 4 show near-perfect clustering (~ 395 correct each), indicating strong separation with the self-tuning graph.
- Class 5 shows partial confusion (149 correct, 225 misclustered) likely due to visual similarity with another class.

Why Clustering Assignment Matrix Values Shift?

1. Cluster labels are arbitrary in unsupervised learning Every time the algorithm runs, it groups the same images together correctly but assigns the group numbers (1, 2, 3, 4, 5) randomly. e.g: So "Forest" might be Cluster 1 in Run 1 and Cluster 4 in Run 2, shifting entire columns in the confusion matrix, even though nothing actually changed in the grouping.
2. Eigen decomposition and KMeans introduce randomness each run Spectral clustering computes eigenvectors internally these can flip in sign or reorder slightly between runs due to mathematical non-uniqueness. KMeans then re-labels clusters from random starting points on top of that. This chain of randomness changes the column order every run, but since ACC/NMI/ARI use optimal label matching before scoring, the performance metrics stay identical.

PCA VARIANCE RETENTION



This graph is called PCA Variance Retention Curve.
It shows:

- X-axis: Number of Principal Components
- Y-axis: Cumulative Variance Explained (%)

Beginning of the graph

- The curve rises very steeply.
- Meaning:
 - ➡ First few principal components capture most of the information.

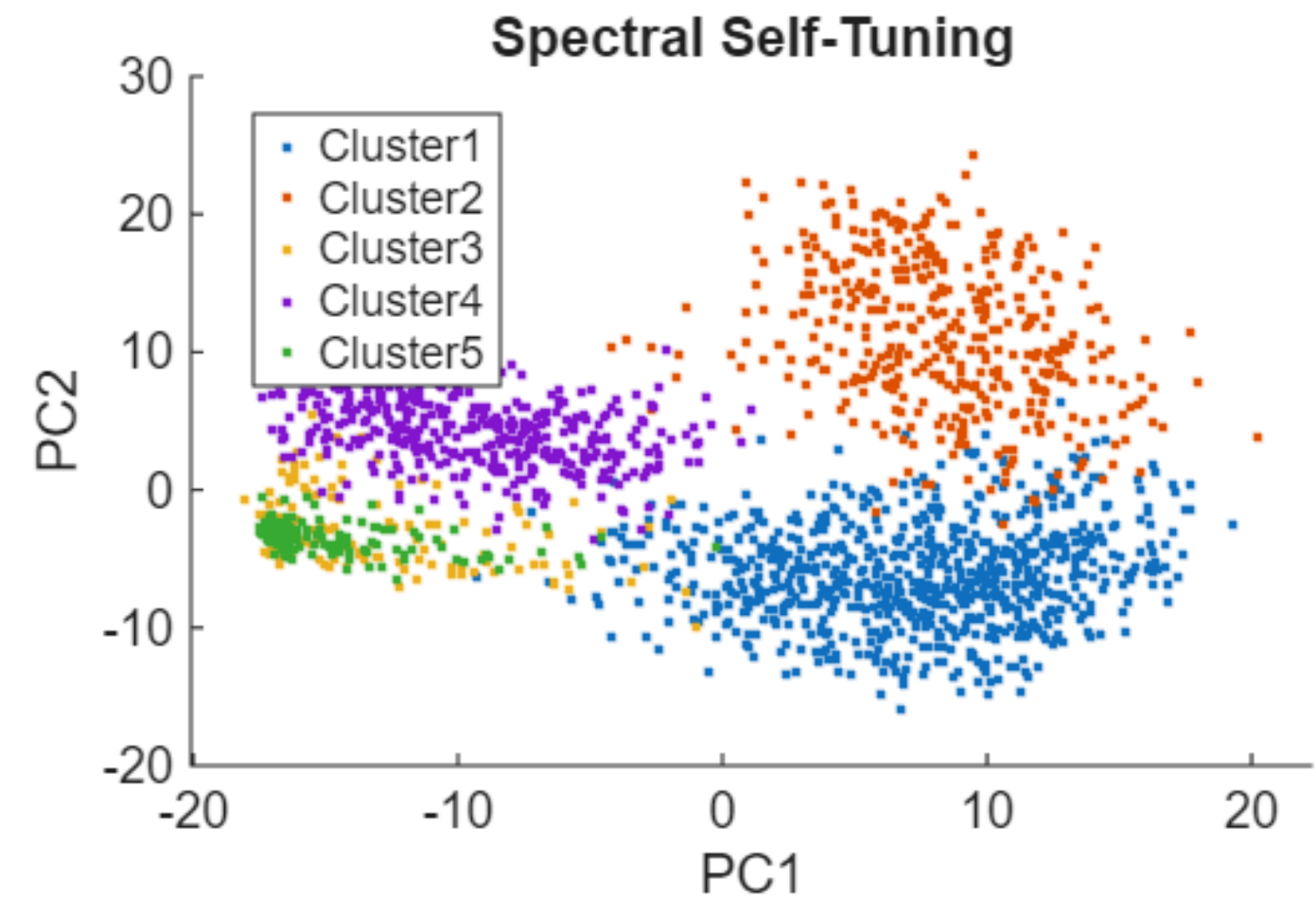
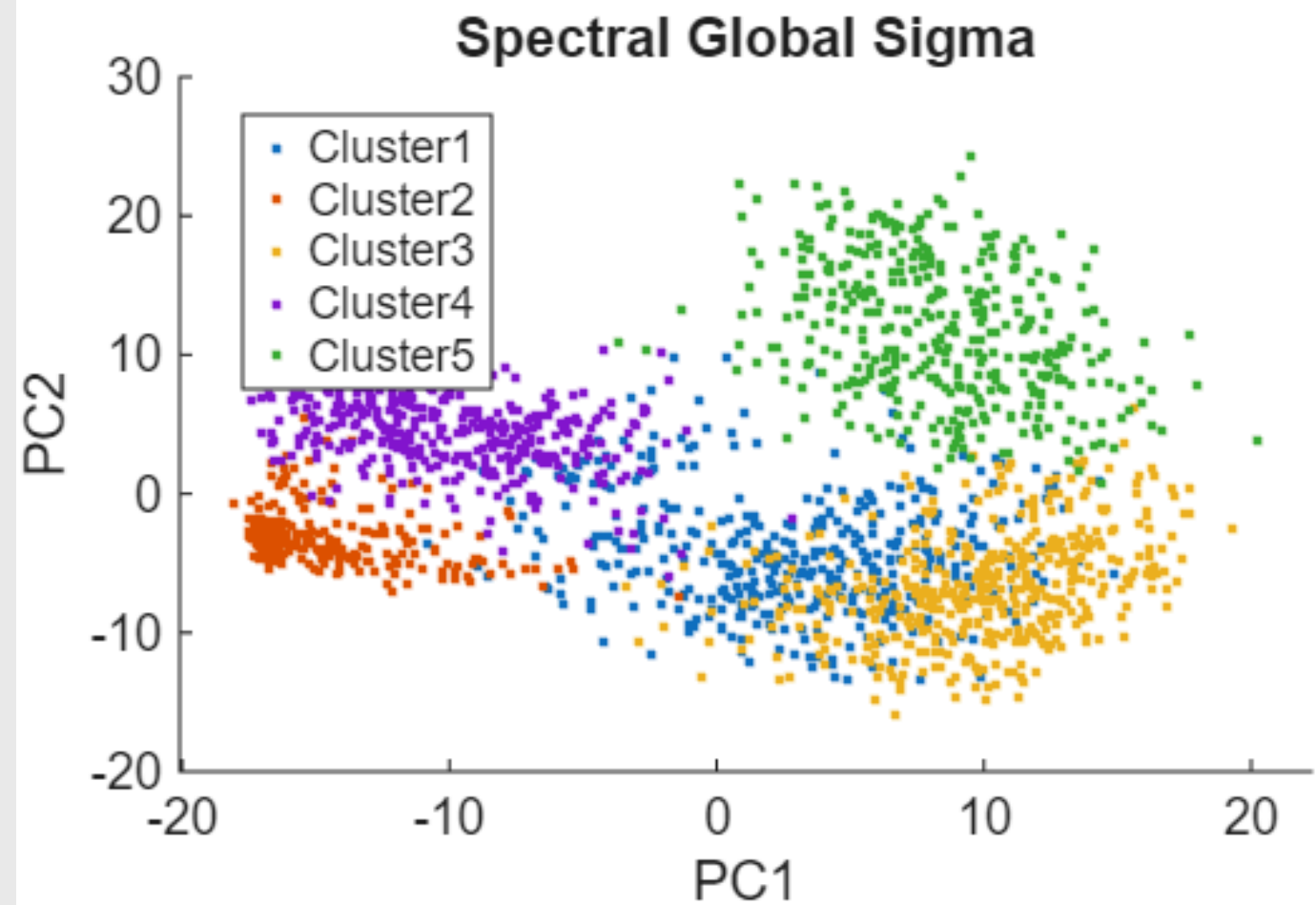
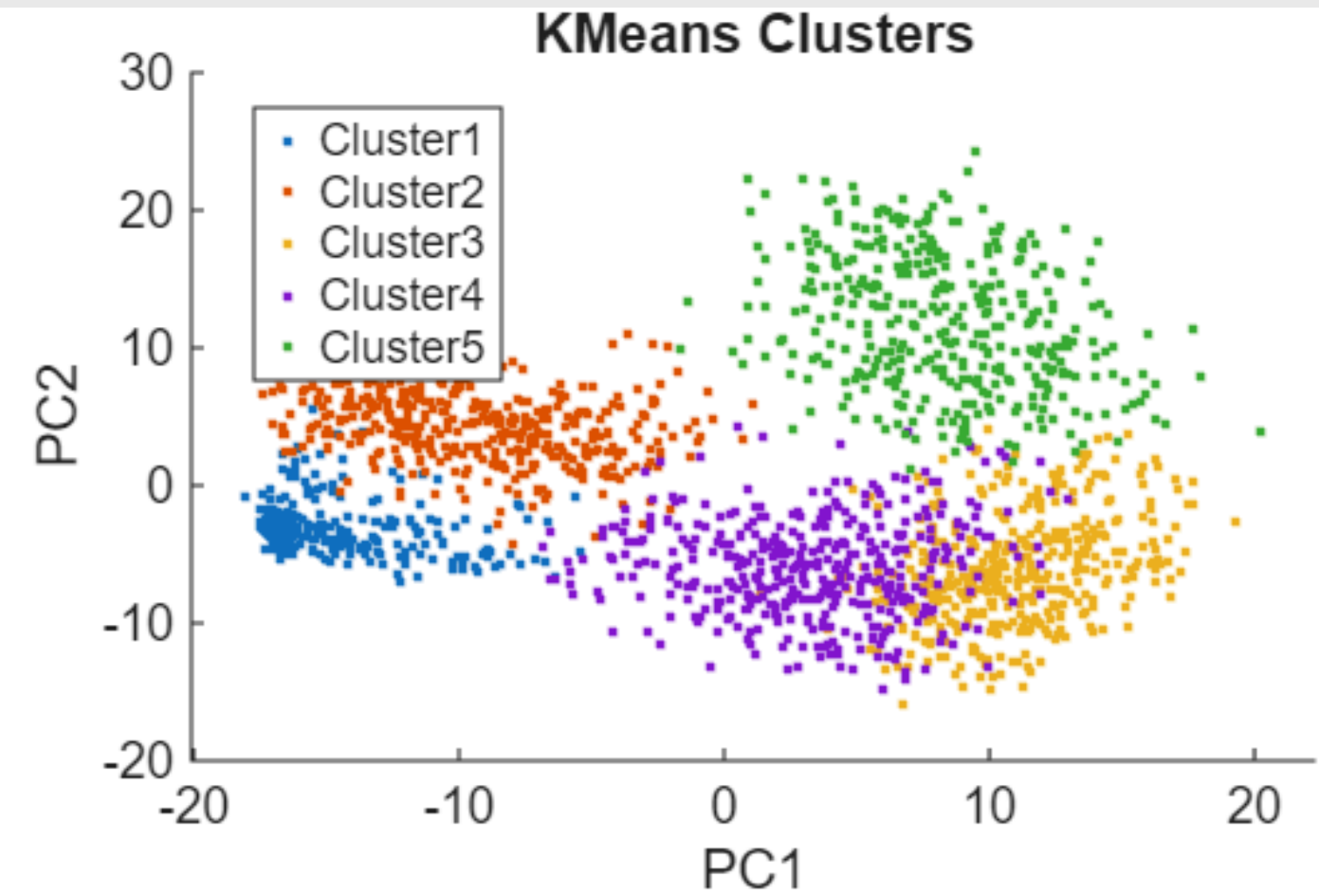
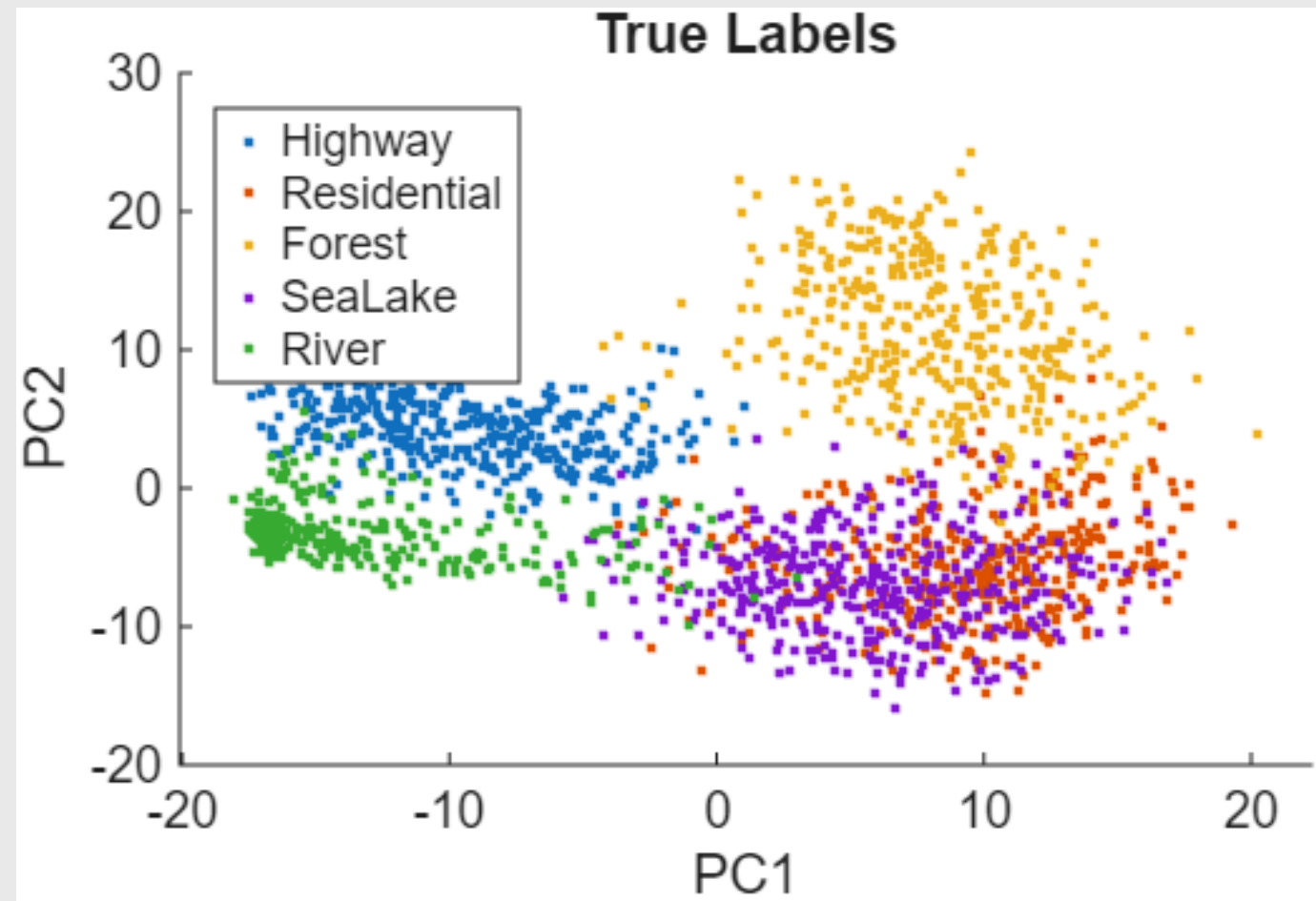
Middle of the graph

- The curve starts flattening.
- Meaning:
 - ➡ Additional components add very little new information.

- **End of graph**
- The curve reaches almost 100% variance.
- Meaning:
 - ➡ Using all components gives full information (same as original dataset).

“ It tells how much information from the original dataset is preserved when we reduce dimensions using PCA (Principal Component Analysis)”

CLUSTERS DISTRIBUTION :



First Plot – True Labels

This plot shows the actual classes in the dataset like Highway, Forest, River, etc.

Each color represents the real category of the data.

PCA Projection

The data originally had many features, so PCA reduced it to 2 dimensions:

PC1 → First principal component

PC2 → Second principal component

This helps us visualize the data easily in 2D.



K-Means Clustering :

K-Means groups the data based on distance between points.

It creates 5 clusters

But the clusters do not perfectly match the real labels.

Reason: K-Means works best for simple round-shaped clusters.

Spectral Clustering (Global Sigma):

This method uses similarity between points instead of just distance.

It builds a graph of relationships between points.

Result:

Clusters are slightly better than K-Means.

Spectral Clustering (Self-Tuning) :

This is an improved version of spectral clustering.

It adjusts similarity based on local data density.

Result:

Clusters match the real labels more closely.

- **"This figure compares true dataset labels with clustering results. PCA is used to visualize the data in 2D, and different clustering algorithms are applied. Spectral self-tuning clustering produces the best grouping because it adapts to local data structure."**

CLUSTERS EVALUATION

METRICS:

Clustering Evaluation Metrics

Metric	Full Form	Mathematical Formula	Range	What It Measures	Interpretation
NMI	Normalized Mutual Information	$NMI(U, V) = \frac{2I(U, V)}{H(U) + H(V)}$	$0 \rightarrow 1$	Shared information between true labels and predicted clusters	1 = perfect clustering, 0 = no relation
ARI	Adjusted Rand Index	$ARI = \frac{RI - Expected(RI)}{Max(RI) - Expected(RI)}$	$-1 \rightarrow 1$	Agreement between pairs of samples in clusters	1 = perfect match, 0 = random clustering
ACC	Clustering Accuracy	$ACC = \frac{1}{N} \sum_{i=1}^N 1(y_i = map(c_i))$	$0 \rightarrow 1$	Percentage of correctly clustered samples after optimal label mapping	1 = all samples correctly clustered

THANK YOU